

Assessing RAG: A Comprehensive Review of Evaluation Frameworks

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Abstract—Retrieval-Augmented Generation (RAG) pipelines, which combine retrieval and generative AI components, are becoming crucial for accurate context-aware text generation in a variety of applications. However, this remains challenging for RAG systems to evaluate due to their complexity and the need to evaluate retrieval and generation components. This survey is conducted to fill the gap in systematic evaluation methodologies by examining existing RAG evaluation frameworks and providing practical advice to improve performance. We compare and evaluate according to metrics such as precision, faithfulness, and adaptability. Our analysis shows that open-source tools focus on transparency and customization but demand some technical knowledge, while the proprietary options focus on integration ease at the cost of flexibility. Major findings show that the granularity of metrics is related to the computational cost and proximity to the domain-specific needs. This work highlights the importance of rigorous evaluation frameworks for RAG systems to achieve practical reliability in real-world applications, thus closing the gap between theoretical benchmarks and real-world deployment.

Index Terms—RAGAS, ARES, COFE-RAG, Haystack, LlamaIndex, Langchain.

I. INTRODUCTION

Retrieval-Augmented Generation (RAG) systems are becoming increasingly important in enhancing Large Language Models' capabilities by leveraging external knowledge sources [1]. However, evaluating these systems is critical due to their modular nature and the challenges in assessing long-form responses. Comprehensive evaluation frameworks like FRAMES and RAGChecker have been proposed to assess RAG systems' performance in factuality, retrieval, and reasoning [2][3]. These frameworks aim to provide a clearer picture of LLM performance in end-to-end RAG scenarios and guide the development of more effective systems. This survey is conducted to fill the gap in systematic evaluation methodologies by examining existing RAG evaluation frameworks and providing practical advice to improve performance [4].

Retrieval-Augmented Generation (RAG) systems combine retrieval and generation components to enhance language models with external knowledge [5][6]. Retrieval-Augmented Generation (RAG) pipelines are two core components combined: A retriever that gets context from external knowledge sources and a generator (usually an LLM) that writes responses based on the retrieved information [1]. Evaluating RAG systems poses challenges due to their hybrid nature and reliance on dynamic knowledge sources [5]. Common evaluation metrics include relevance, accuracy, and faithfulness for both retrieval and generation components [5][6]. Recent

frameworks like RAGAS propose reference-free evaluation metrics [6], while RAGBench offers a large-scale benchmark dataset and the TRACe evaluation framework [7]. RAGEval introduces scenario-specific evaluation dataset generation and novel metrics such as Completeness, Hallucination, and Irrelevance [8]. These approaches aim to address the limitations of current benchmarks and advance RAG evaluation methods, enabling more comprehensive assessment of RAG systems across various domains and applications [5][7][8].

II. RAGAS (RETRIEVAL-AUGMENTED GENERATION ASSESSMENT)

Traditionally, we use two main methods to assess the performance of Retrieval-Augmented Generation (RAG) systems. We start by evaluating the model's language generating abilities using measures like perplexity or the probability of the generated text, which have been shown in studies like [9]. Besides that, we use question-answering benchmarks, a technique demonstrated by [10], to compare the model's factual correctness against carefully selected, human-labelled datasets.

Understanding the drawbacks of conventional RAG evaluation, especially its dependence on data with human labels, the RAGAS framework provides a different method. Established, annotations are no longer necessary because this methodology introduces reference-free indicators. RAGAS uses:

- **Faithfulness:** This measure evaluates how well the generated response is rooted in the context that was retrieved, so confirming that the model's output is backed up by the data that it accessed. Existing factuality estimation methods, as those used in BARTScore and FactScore [11], serve as the basis for this assessment.

- **Context Relevance:** Efficiency depends on the retrieved information being both relevant and succinct. As explained by [14], RAGAS assesses context relevance to ensure that the retrieved passages are targeted, minimizing the presence of unnecessary material and so lowering computational cost.

- **Answer Relevance:** As demonstrated in [15], RAGAS uses embedding-based similarity assessments to analyze whether the generated response directly answers the query submitted by the user. This efficiently measures how closely the response relates to the question's intent.

Without the limitations of needing manually produced, ground-truth annotations, these automated measures are made to closely resemble human evaluations, enabling a more effective and flexible evaluation procedure.

III. ARES (AUTOMATED EVALUATION FRAMEWORK FOR RETRIEVAL AUGMENTED GENERATION SYSTEMS)

ARES expands on earlier initiatives like RAGAS and EXAM, which assessed RAG pipelines using either heuristic prompting or QA-based simulations. In contrast to these methods, ARES employs a multi-stage pipeline that refines LLM judges (such as DeBERTa-v3-Large) on artificially generated question-passage-answer triplets, enabling targeted classification along three axes: Answer Relevance, Context Relevance, and Answer Faithfulness. The limitations of the entire assessment metrics, which frequently mask flaws, are addressed by this tri-metric approach, which provides a detailed and modular study of RAG processes.

ARES leverages PPI, a statistical technique that generates confidence intervals on predictions made by LLM judges on huge unlabelled sets using a small human-annotated validation set (about 150 cases), to increase the trustworthiness of its evaluations. This makes ARES both cost-effective and data-efficient by enabling it to retain high accuracy while significantly lowering the requirement for manual labelling. Indeed, ARES uses 78% fewer annotations and performs 14.4–59.9% better than RAGAS in evaluation accuracy across several datasets (KILT, SuperGLUE, and AIS).

Across domains and datasets, including knowledge-driven tasks like NQ, HotpotQA, FEVER, and MultiRC, ARES exhibits strong generalization. Both actual and pseudo RAG systems are efficiently ranked, and it exhibit good agreement with benchmarks that have been labelled by humans. Crucially, PPI helps to attenuate performance declines in cross-domain transfer, which occurs when a judge trained for one domain (like NQ) still performs extremely well in another (like ReCoRD or FEVER) as mentioned by [17].

IV. CoFE-RAG (COMPREHENSIVE FULL CHAIN EVALUATION FOR RETRIEVAL AUGMENTED GENERATION)

The CoFE-RAG framework offers a thorough full-chain evaluation technique for Retrieval-Augmented Generation (RAG) systems. It was first presented by [14]. Traditional RAG evaluation techniques have a number of drawbacks, such as unpredictable retrieval evaluations, limited data diversity, and trouble pinpointing specific RAG pipeline failure sites. By implementing multi-granularity keyword assessments, CoFE-RAG resolves these problems and makes it possible to evaluate the retrieval and reranking phases more precisely and systematically without depending on manually annotated golden chunks. The system reduces reliance on time-consuming annotation procedures while improving retrieval performance evaluation by combining coarse-grained and fine-grained keywords. CoFE-RAG's relevance for assessing RAG models in practical settings is further strengthened by the inclusion of a benchmark dataset that spans a variety of file types and query kinds.

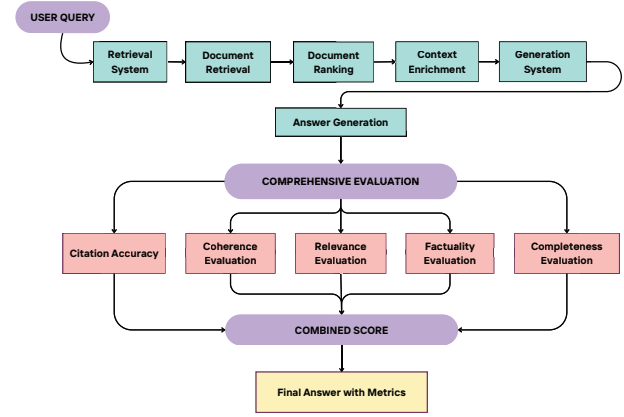


Fig. 1: CoFE-RAG Framework

Implementing a comprehensive evaluation framework throughout the RAG pipeline is how CoFE-RAG (Comprehensive Full-chain Evaluation for Retrieval-Augmented Generation) operates. The retrieval system above is prompted to retrieve pertinent materials from a knowledge base by a user query at the start of the procedure. Before being sent to the generation system, these documents are first sorted according to their relevancy and then enhanced with contextual data. The query and the information obtained are both used by the model to generate an answer. The thorough evaluation layer of CoFE-RAG sets it apart from conventional RAG techniques by evaluating the generated response on several criteria, such as completeness, coherence, factual accuracy, relevance to the inquiry, and appropriate source reference. The sum of these separate criteria yields an overall rating that makes the response's quality transparent. CoFE-RAG seeks to provide more dependable and traceable AI-generated responses by assessing the complete chain from retrieval to generation and detecting possible flaws in certain pipeline components [12].

V. DEEPEVAL

An open-source framework called DeepEval was created especially to assess Retrieval-Augmented Generation (RAG) pipelines. It offers a methodical way to examine the retrieval and generation elements of these systems. It makes sure that RAG models produce responses based on retrieved context by measuring important factors including retrieval accuracy, response relevance, and contextual consistency. The framework is appropriate for practical applications where retrieval accuracy and generation quality need to be regularly checked because it interfaces easily with LangChain and LlamaIndex [13]. Furthermore, DeepEval facilitates the creation of synthetic datasets and lets users provide unique assessment criteria, providing flexibility for domain-specific applications.

An evaluation of pen-source large language models (LLMs) in the field of enterprise-specific RAG domains was conducted and utilized in a study conducted by [15] where DeepEval was made use of to achieve a clear comparison between retrieval

quality and generative responses. The authors point out that DeepEval offers contextual precision, contextual recall, and contextual relevancy as crucial measures to help identify situations in which retrieval is unable to bring in accurate supporting documents. Therefore, DeepEval turns out to be a sophisticated tool for RAG evaluation enabling hyperparameter optimization, hallucination detection, and enhanced retrievals.

A. Applications of DeepEval in RAG Evaluation

The study covers several important uses for DeepEval in assessing RAG pipelines, with a focus on enterprise AI, automated testing, and retrieval accuracy benchmarking. These uses demonstrate how DeepEval can offer standardised, repeatable, and scalable assessments in a variety of fields.

B. Automating RAG Model Evaluation

The automated assessment of RAG models, which eliminates the need for manual verification of retrieval and generation accuracy, is one of DeepEval's main uses. The study emphasises how DeepEval and RAGAS were utilised to expedite the evaluation process by substituting automated scoring based on retrieval relevance, answer faithfulness, and contextual precision for manual test case classification. Large-scale RAG implementation evaluation takes a lot less time and effort thanks to this automation, which makes it the go-to option for AI researchers and engineers.

VI. RAGEVAL

Production-level RAG systems have requirements that are not well-handled in standard machine-learning development workflows. As [17] pointed out, the very stochastic nature of Large Language Models, paired with rather complex multi-component RAG architectures, makes it explicitly required to have specific tools for its successful development, assessment, and monitoring. Modern evaluation platforms provide a holistic solution, providing a bridge between software prototyping and deploying in production, and a unified platform for addressing the full life cycle of RAG system development. We can finally see the special place it takes in treating evaluation as an integrated whole across the entire lifecycle, from early experiments through to production monitoring and improvement.

A. Integrated Development Model

Contemporary RAG evaluation platforms are based on the principle that sound (good) RAG development needs the cross-linking of several processes that have been considered as separate traditionally. The platform shifts the balance toward production and confronts the core challenge highlighted by recent research [15]: how to ensure that quality and reliability from proof-of-concept (PoC) demos do not get lost as PoC gets transformed to production applications.

B. Continuous Integration of the Evaluation

Advanced frameworks do not take evaluation information as a monolithic phase of any development process; instead, evaluation capability is woven into each phase and made available at each stage of the process for continuous consideration and iterative development. Continuity from Production to Development The platform also guarantees that evaluation methods and metrics used in development are the same as those in production monitoring, as well as enabling seamless transitions.

VII. MACHINE LEARNING OPERATIONS (MLOPS) FRAMEWORKS

The use of Large Language Model evaluation as an integrated component of standard Machine Learning Operations (MLOps) frameworks is a monumental step in the field of AI system creation and deployment. As pointed out by recent research [21], the rise of Large Language Model Operations (LLMOps) as a distinct category within MLOps aims to (i) specifically deal with the need of MLOps for LLMs, such as high resource consumption, the need of continuous training, complex evaluation, and ethical challenges such as fairness and bias. Contemporary LLM evaluation frameworks mitigate these issues by building on the built-in experiment tracking, model versioning, and lifecycle management features of the platform, and tailoring them to the specific needs of LLM systems, specifically Query-Augmented Generation systems.

End-to-end tracing of event-driven workflow execution, also known as Workflow Observability, provides a comprehensive view of a system's internal stages and decisions. This is complemented by Component-Specific Monitoring, which tracks the performance of individual parts of the workflow, such as retrieval methods, context processing, and generation. Together, these monitoring efforts support Performance Tuning, allowing for the analysis of different query strategies and configuration parameters to identify the most efficient system settings.

VIII. CONTINUOUS EVALUATION FRAMEWORK

Using Large Language Models (LLMs) in production has special challenges to keep performance and quality stable. These cost and operational risks are detected and mitigated by contemporary continuous evaluation frameworks, which offer holistic data-driven evaluation toolsets tailored specifically for the continuous evaluation of LLM-powered apps in production [16].

A. Data-Driven Evaluation Paradigm

There is a move from periodic, manual evaluation practices to systematic, data-driven evaluation methodologies by using modern continuous evaluation frameworks [16]. By using production data ingestion, historical performance analysis, and user-interaction logs, the framework offers a complete view of LLM performance than standardized benchmarking methods. They take an empirical approach, with data at heart, to evaluate performance in real-world settings that cut through the noise and vagaries of full-scale environments [8]. The framework's

flexibility to include a wide range of data sources yields a general understanding of the real user experience and system behaviour across various operational scenarios.

B. Continuous Monitoring and Real-Time Assessment

The platform's real-time monitoring engine introduces advanced performance drift detection and quality regression monitoring capabilities [18]. This live estimation pipeline allows organizations to detect performance degradation as soon as it occurs, assisting proactive maintenance and quality control methods. The feature has predefined rules that are capable of being managed to meet specific organisational needs and quality standards [17]. This feature is essential to meet production SLAs and provide a consistent user experience.

IX. LLAMAINDEX

Llamaindex is now standing as a prominent framework which is being widely tractioned because of its ability to conceptualize Retrieval-Augmented Generation (RAG) pipelines in simplicity, modularity and integration capabilities. It facilitates the construction of end-to-end RAG systems by utilizing a limited number of code lines, therefore resulting in an accessible way for beginners while simultaneously providing advanced personalization for research-based applications. The primary strength of the program arises from it facilitating indexing, chunking, retrieval and query processing which offers built-in parsers such as SimpleDirectoryReader and LlamaParse. Additionally, as a support for embedding models and retriever parameter tuning. Nonetheless, in spite of its efficiency, Llamaindex has constraints: its documentation is often incomplete or tough to navigate, which may pressure the researchers to improvise explanations. The majority of the codes given are either exceedingly simple or overly progressive, which leads to gaps in applied guidance [19].

X. LANGCHAIN

While focusing primarily on RAG, large language models have widely adopted the LangChain framework for building applications. The integration of retrieval, generation, and reasoning components provides developers with a modular design to help reduce complexity when constructing pipelines. LangChain consists of several modules, like LangGraph, which uses graph-based orchestration to support multi-agent applications with memory of prior interactions, an API layer offered by LangServe facilitates the scaling and serving of LangChain applications in production settings, and LangSmith, which ensures transparency and constant enhancement through debugging, monitoring, and assessment of RAG pipelines throughout the application process [20].

XI. HAYSTACK

The paper Document Haystacks: Vision-Language Reasoning Over Piles of 1000+ Documents by Wang et al. (2024) introduces two document haystack benchmarks, DocHaystack and InfoHaystack, designed to evaluate Large Multimodal Model (LMM) performance on large-scale visual document

retrieval and understanding tasks involving over 1000 documents. The authors propose V-RAG, a vision-centric retrieval-augmented generation framework that incorporates multiple multimodal vision encoders and a dedicated question-document relevance module, achieving 9% and 11% improvements in Recall@1 on challenging benchmarks. The key contributions include the introduction of large-scale multimodal retrieval benchmarks, the V-RAG framework with ensemble encoders and relevance filtering, and a comprehensive evaluation of existing LMMs on large-scale retrieval tasks, demonstrating significant performance gains through specialized retrieval architectures.

XII. IMPLEMENTATION

A. Llama Index Framework

1) *System Implementation:* Dependencies (llama-index, pinecone, gradio) are in requirements.txt. The app sets up Pinecone, loads Paul Graham's essay via SimpleDirectoryReader, chunks and embeds it, then indexes to Pinecone using VectorStoreIndex.from_documents(). A query engine processes questions via semantic search and LLM synthesis. A Gradio UI is deployed on Hugging Face Spaces.

2) *Test Case Results:* Test Results: Querying "what does Paul Graham say about startups?" returned a correct response about the Y Combinator "batch model," showing effective retrieval and answer synthesis.

B. LangChain Framework

1) *System Implementation:* Requirements include LangChain modules for Pinecone and OpenAI. The essay is loaded and split with CharacterTextSplitter, embedded with OpenAIEmbeddings, then stored in Pinecone. RetrievalQA.fromchaintype() powers the RAG chain. A simple Gradio interface is used.

2) *Test Case Results:* Test Results: The same test query produced an identical answer as llamaIndex, confirming LangChain's capability for accurate retrieval and generation.

C. Haystack Framework

1) *System Implementation:* Haystack's pipeline design uses separate indexing and query pipelines. The indexing pipeline (TextFileToDocument, DocumentSplitter, OpenAIDocumentEmbedder, DocumentWriter) processes and stores the essay in Pinecone. The query pipeline (OpenAITextEmbedder, PineconeEmbeddingRetriever, PromptBuilder, OpenAIGenerator) retrieves context and generates answers.

2) *Test Case Results:* Again, the query returned a consistent answer about the "batch model," demonstrating Haystack's modular and reliable RAG construction.

XIII. CONCLUSION

This thorough analysis has looked at the state of Retrieval-Augmented Generation assessment frameworks today, emphasizing both the noteworthy advancements and the obstacles still present. While frameworks such as RAGAS, ARES,

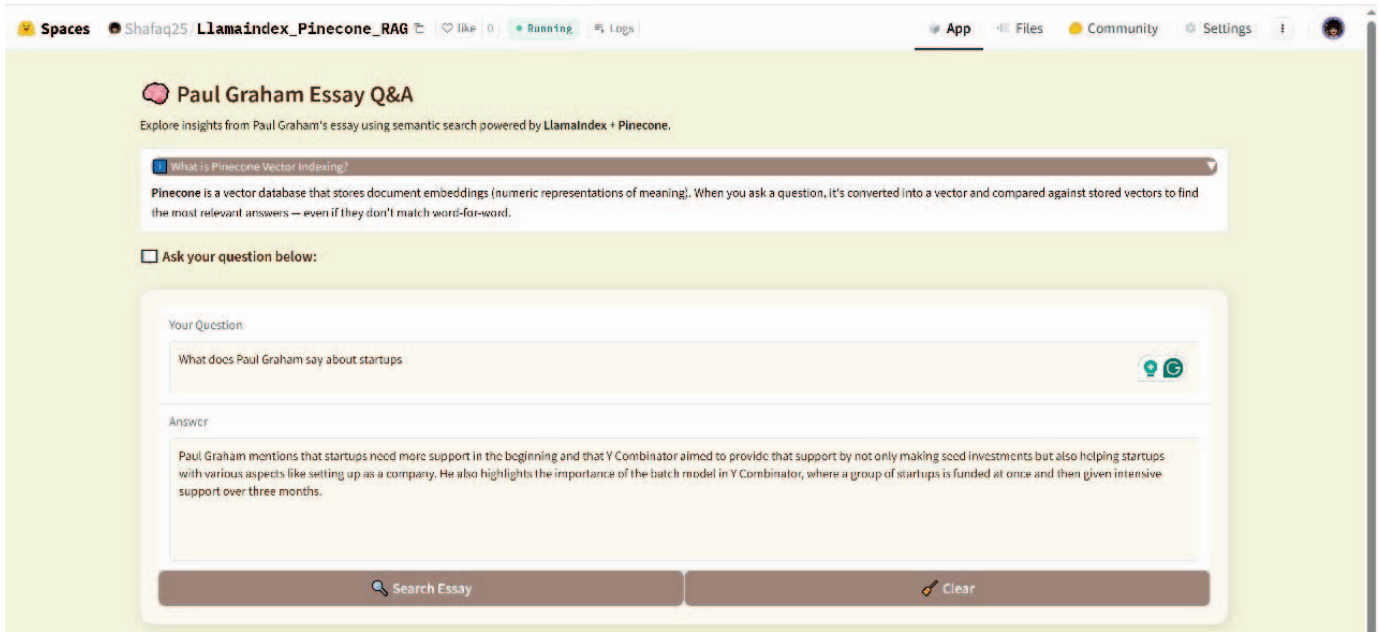


Fig. 2: Llama Index User Interface

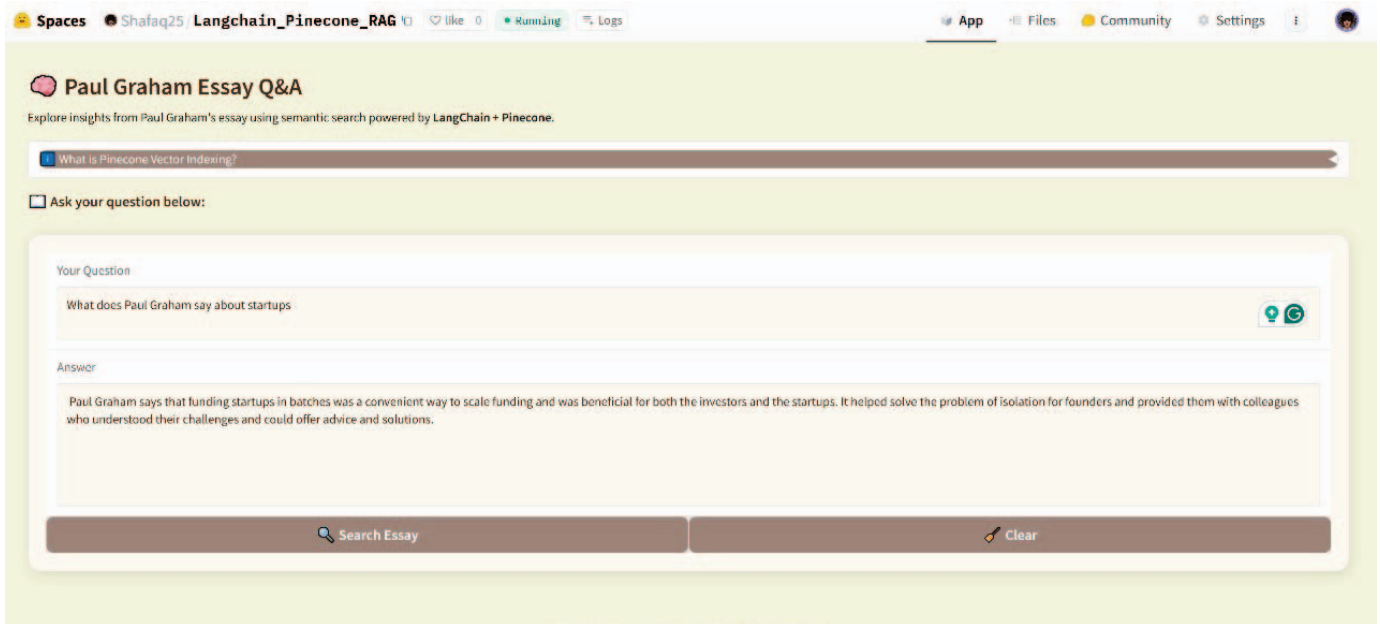


Fig. 3: LangChain User Interface

CoFE-RAG, DeepEval, and RAGEval provide advanced approaches for evaluating RAG systems in a variety of ways, such as faithfulness, context relevance, answer quality, and factual accuracy, our analysis shows that no single framework offers a comprehensive solution.

The unique use cases and requirements need to be aligned

with the tough decisions that practitioners and researchers face, which include options between specialized component-level assessment and holistic system evaluation, automated reference-free evaluation and human-annotated benchmarks, and between open-source flexibility and ease of proprietary integration.

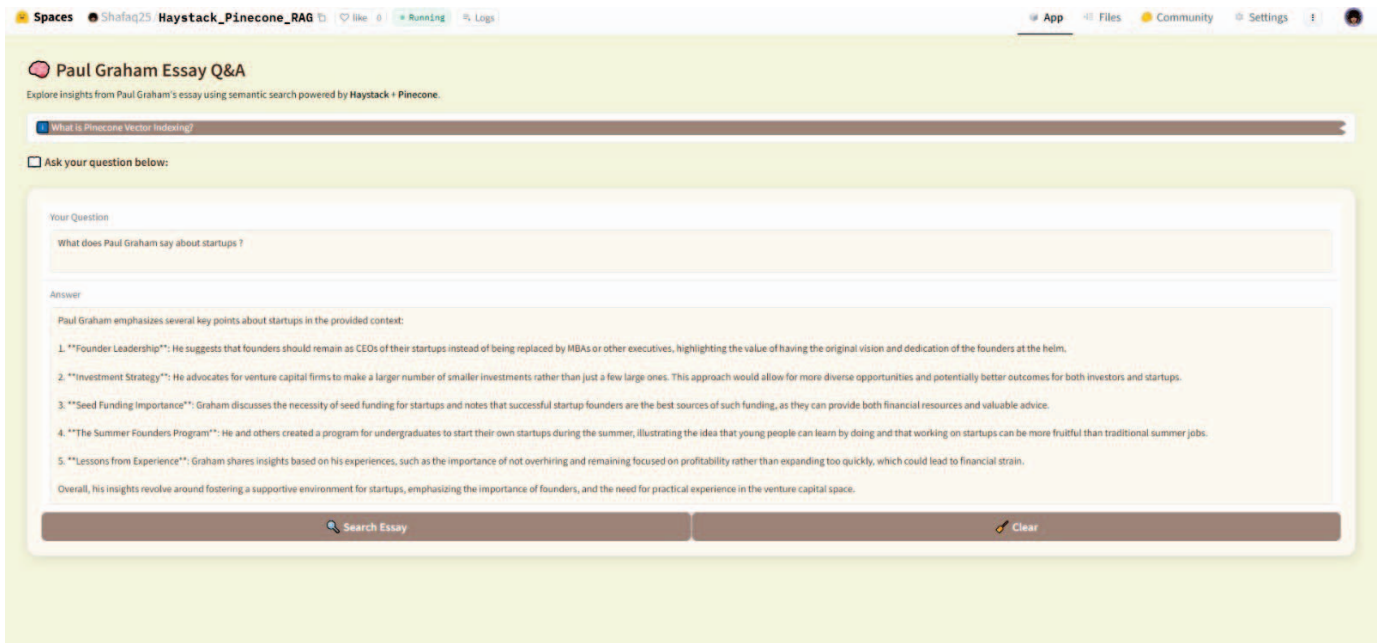


Fig. 4: Haystack User Interface

The creation of more thorough, consistent, and reliable assessment techniques will be essential to guaranteeing the dependability, security, and efficacy of RAG systems in practical implementations as they continue to expand and find uses in a variety of fields. In order to stay ahead of the quick developments in retrieval-augmented generation technologies, future research should concentrate on establishing standardized standards, resolving cross-domain generalization issues, and devising more effective assessment methods.

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